Co-Design Rankings Do Not Survive Realization: An 18-Morphology, 4-Environment Study

Author One, Author Two, Author Three
Creative Machines Lab, Columbia University, New York, NY, USA
{author1, author2, author3}@columbia.edu

Abstract*—Learning-based co-design methods score candidate robot bodies in a simplified simulator and return a ranked list. We ask how much of that ranking survives when the designs are made buildable. We take 18 morphologies of a modular icosahedral legged robot, sampled from our own evolutionary co-design space, and train each with the same PPO recipe in a replica of the abstract simulator used by the search and in a fabricable simulator calibrated against CAD models and motor datasheets. The two rankings are uncorrelated ($\rho_s = +0.11$, $p = 0.66$; mean rank change 5.9 of 18): the best abstract design drops to 7th, and a design that barely moved in the abstract simulator (0.54 m, 17th) becomes 2nd at 10.8 m. The reshuffle is robust to the training seed ($P < 0.01$ under resampling), to reward re-weighting and added mass ($\rho_s \geq 0.77$) and to halving the training budget ($\rho_s \geq 0.90$), and it is caused mainly by the changes in geometry, DoF allocation, rest pose and mass rather than by the actuator model ($\rho_s = -0.30$ vs. $+0.61$). Changing the terrain at fixed realization, in contrast, largely preserves the ranking (rubble $\rho_s = 0.90$, ice 0.72); only stepping stones reshuffle it as much. We trace the reshuffle to a shell-sliding exploit, to rest poses that penetrate or float above the ground, and to a leg-count advantage that exists only in the abstract simulator. Simulator fidelity should therefore be part of the co-design loop rather than a step after it.

I. Introduction

Most learning-based co-design methods, whether they propose morphologies generatively or improve them iteratively by RL, optimization or evolution [1]–[6], produce the same kind of output: a ranked shortlist of bodies, each scored by a policy trained in a deliberately simplified simulator. Our own prior work follows this pattern: ICOS is an orientation-free Platonic shell in the tradition of polyhedral robots [7]–[11], and we searched its leg placements with a speciated evolutionary algorithm in MuJoCo MJX [12] (seven DoF-partition species). Here we do not repeat that search; we sample 18 bodies from its design space and ask what happens to their ranking when each body becomes a machine printed and assembled from Dynamixel XM430-W350 servos.

II. Experimental Setup

Robot and morphologies. ICOS is a 20-face icosahedral shell (160 mm circumdiameter) to which identical 3-DoF leg modules are bolted on chosen faces. All 20 faces are equivalent, so a morphology is a subset of faces (60,249 possibilities for 3–6 legs), and its symmetry follows from where and how the legs are mounted. The realized quadruped weighs 2.20 kg, stands 134 mm tall and uses 12 servos. We study 4 hand-designed baselines and 14 generation-0 samples, two per

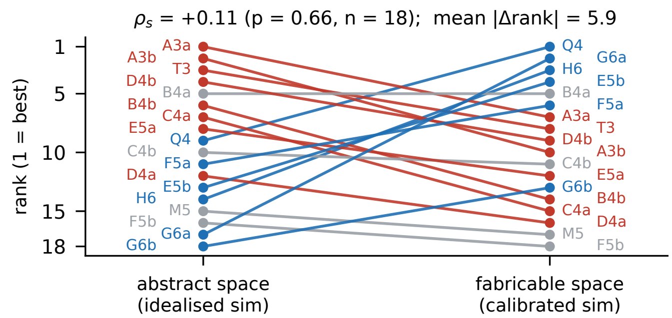

Fig. 1. Flat-ground ranking of the 18 morphologies in the abstract (left) and fabricable (right) space. Blue lines rise, red lines fall, grey lines move at most 2 ranks. Q4/T3/M5/H6: hand-designed quadruped, tripod, mixed, hexapod; A–G: the seven DoF-partition species, two generation-0 samples (a, b) each; the digit is the leg count.

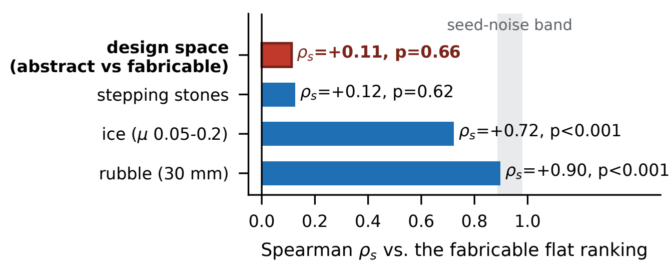

Fig. 2. Spearman rank correlation of each ranking with the fabricable flat-ground ranking. Shaded: the 5–95% range that seed noise alone produces at the median measured CV.

species; the set is a stratified sample of the search space, not of its best designs.

Training. All runs use the same recipe: Genesis [13] GPU simulation for both design spaces (the abstract space re-implements the MJX models used by the search), 4096 parallel environments, PPO (rsl-rl [14]), 2000 iterations (197M steps), 20 s episodes, and 50 Hz control on top of 250 Hz physics. The reward is $r = 2.5\,\Delta d + 0.5\,\text{alive} - 0.01\|a\|^2$; in the fabricable space a thermal penalty of -0.1 is added, except in the ice runs (Sec. IV). The only other difference between the two spaces is the entropy coefficient (0.01 abstract, 0.003 fabricable). We report the mean displacement over 4096 rollouts (never the best of several automatic retries) and an *honest* variant that truncates each rollout at its first fall (Sec. III-C).

Realization. Five things change between the abstract and the fabricable model. (1) The freely partitioned 12-DoF budget (2–5 DoF per leg) becomes a fixed printable 3-DoF module, so DoF $= 3\times$legs; a $(4, 4, 4)$ design, for example, cannot be built from one module. (2) Sphere collision proxies become a convex-hull body mesh with motor cylinders and skirt rings. (3) The straight-leg rest pose, which the real robot cannot reach, becomes a solved static equilibrium ($\Delta z < 25$ mm, no penetration). (4) The ideal position servo becomes a torque PD controller limited by a torque–speed envelope (4.1 N·m stall, 4.8 rad/s no-load), with 0–20 ms latency, sensor noise and the thermal penalty. (5) Lumped masses become CAD part masses at their centroids, so mass now scales with leg count (1.8–3.1 kg and 9–18 servos for 3–6 legs). Changes (2)–(5) follow standard sim-to-real practice [15], [16], with parameters taken from CAD over seven hardware generations.

III. Results

A. Realization reshuffles the ranking

The two flat-ground rankings (Fig. 1) are statistically unrelated ($\rho_s = +0.11$, $p = 0.66$); the mean rank change is 5.9 of 18, close to the 6.0 expected for independent rankings. The best abstract design (3 legs, 17.5 m) finishes 7th at 5.53 m in the fabricable space, and two designs that barely moved in the abstract space (0.54 and 0.13 m) finish 2nd (10.8 m) and 13th.

Seed noise does not explain this result. Retraining 6 morphologies with 3 seeds each on fabricable flat ground gives a median coefficient of variation of 0.13 (range 0.03–0.63) and seed-to-seed rank agreement of $\rho_s = 0.77$–0.89. Under the null hypothesis that only seed noise moves an otherwise unchanged ranking, 20,000 resamples give $P(\rho_s \le 0.11) < 10^{-4}$ at the median noise level and 0.006 at the highest.

Reward weights and training budget do not explain it either. Raising the distance weight of the abstract reward from 1.0 to 2.5, or adding 57% realistic part mass, changes the abstract ranking little ($\rho_s = 0.77$–0.97 among four variants) without moving it toward the fabricable one ($\rho_s \le 0.11$); the rankings at half the training budget already match the final ones ($\rho_s \ge 0.90$) and already disagree across spaces ($\rho_s = 0.21$).

To separate the actuator model from the other changes, we retrained the 18 fabricable bodies with the ideal servos of the abstract space. The resulting ranking is unrelated to the abstract ranking ($\rho_s = -0.30$, $p = 0.23$) but close to the fully calibrated one ($\rho_s = +0.61$, $p = 0.007$). The changes in geometry, DoF allocation, rest pose and mass therefore account for most of the reshuffle. The actuator model perturbs the ranking further and unevenly (the best design loses 13%, one design loses 55%, another gains 2.5×), which is why 0.61 lies below the seed-noise band at median noise (0.89–0.98).

B. The task environment mostly does not

With the fabricable model fixed, we retrained all 18 morphologies on quantized fractal rubble (30 mm), ice ($\mu \in [0.05, 0.2]$) and stepping stones (0.25 m stones, 50 mm gaps over a 0.5 m drop). Rank correlation with the flat-ground ranking remains high (Fig. 2): $\rho_s = 0.90$ for rubble, inside the median-noise seed band, and 0.72 for ice, below it (both $p < 0.001$; the ice runs lack the thermal term, Sec. IV). Only stepping stones reshuffle the ranking as much as realization does ($\rho_s = 0.12$, $p = 0.62$). There, 15 of 18 morphologies learn to survive (standing still is a valid strategy), but only 4 of 18 travel more than 1 m (median 0.49 m).

C. Why the ranking changes

The reshuffle is not a scoring artifact: re-scoring every rollout at its first fall leaves both rankings unchanged (no abstract policy ever falls; $\rho_s = 0.94$ between the two scorings and still 0.07 against the fabricable ranking; 17 of 18 fabricable policies agree within 5%). We find three causes. *(i) Simulation exploits disappear.* The tripod, 3rd in the abstract space with 11.1 m, achieved this by sliding its smooth spherical shell along the ground, which sphere collision proxies allow and a convex-hull mesh does not. In the fabricable space it walks instead and reaches 5.46 m (8th). *(ii) The rest pose matters.* Of the 18 original home keyframes, 6 penetrated the ground and 3 floated above it. Re-solving them (morphology and reward unchanged) changed flat-ground displacement from 5.33 to 10.84 m and from 5.60 to 10.09 m for two designs and from 8.44 to 5.79 m for a third; all results here use the nine re-solved runs. *(iii) The abstract space rewards a trade-off the hardware cannot make.* In the abstract space, displacement is predicted by leg count: fewer, longer legs win ($\rho_s = -0.89$). After realization this relation disappears ($\rho_s = +0.07$), because the fixed module replaces the trade-off between legs and DoF with one between legs and mass. Mirror symmetry of the leg placement, the leading hypothesis of our evolutionary search, shows no detectable relation to displacement in either space at this sample size ($|\rho_s| \le 0.40$, $p \ge 0.10$); only the tripod has a mirror-symmetric placement at all.

D. Advantage depends on the environment

No morphology dominates across environments. On ice, median retention (environment / flat displacement) rises with leg count (0.33, 0.81, 1.06, 0.88 for 3–6 legs; possibly inflated by the missing thermal term); rubble costs little (0.93) and stepping stones almost everything (0.12).

IV. Discussion

A shortlist scored in an abstract simulator says little about the order of the same designs after realization, but it does transfer across tasks. Iterative search and zero-shot generation share this scoring step, so simulator fidelity belongs inside the co-design objective rather than after it.

Limitations. Runs outside the 6×3 replication use a single seed. The main result survives the highest measured seed noise ($P = 0.006$); the finer comparisons (actuator model, ice) hold only at the median noise level, since the worst-case band (0.22–0.82) contains both 0.61 and 0.72. Two reward inconsistencies remain: the ice runs omit the thermal term, so part of the ice result may reflect reward rather than terrain, and the stepping-stone runs of the four baselines carried two extra penalty terms (re-runs are in progress). In the servo experiment, geometry, DoF allocation, rest pose and mass remain confounded. The abstract space is a Genesis replica of the MJX search environment, not yet checked against it, and we have no hardware results yet.

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